ATTACHMENT 1

Item Code	: 204600 - 01 draft 23 June 2025
PM Code	: Ins.P RII
Measurement(mm) ± 2mm	: 260(L) x 110(W)
CURSOR (ATTACHMENT 1)	: T016 - T029
CURSOR (ATTACHMENT 2)	: B062 - B075
Colour of Artwork	: K 100
Source of Artwork	: MAC 1 / CC

[redacted]

YSP RASITOL INJECTION 10mg/ml

Ingredient(s):
Each ml contains:
Furosemide..10mg

Pharmacology (Summary of Pharmacodynamic and Pharmacokinetic):
Pharmacodynamic:
1. Furosemide acts along the entire nephron with the exception of the distal exchange site. The main effect is on the ascending limb of the loop of Henle with a complex effect on renal circulation. Blood-flow is diverted from the juxta-medullary region to the outer cortex.
2. The principle renal action of furosemide is to inhibit active chloride transport in the thick ascending limb. Re-absorption of sodium chloride from the nephron is reduced and a hypotonic or isotonic urine produced.
3. Prostaglandin (PG) biosynthesis and the renin-angiotensin system are affected by furosemide administration and furosemide alters the renal permeability of the glomerulus to serum proteins.

Pharmacokinetic:
1. Furosemide is a weak carboxylic acid which exists mainly in the dissociated form in the gastrointestinal tract. Furosemide is rapidly but incompletely absorbed (60-70%) on oral administration and its effect is largely over within 4 hours. The optimal absorption site is the upper duodenum at pH 5.0. Regardless of route of administration 69-97% of activity from a radio-labelled dose is excreted in the first 4 hours after the drug is given. Furosemide is bound to plasma albumin and little biotransformation takes place.
2. Furosemide is mainly eliminated via the kidneys (80-90%); a small fraction of the dose undergoes biliary elimination and 10-15% of the activity can be recovered from the faeces.

<u>In renal/ hepatic impairment</u>
1. Where liver disease is present, biliary elimination is reduced up to 50%. Renal impairment has little effect on the elimination rate of furosemide, but less than 20% residual renal function increases the elimination time.

<u>The elderly</u>
1. The elimination of furosemide is delayed in the elderly where a certain degree of renal impairment is present.

<u>New born</u>
1. A sustained diuretic effect is seen in the newborn, possibly due to immature tubular function.

Indication(s):
Furosemide is indicated in the treatment of oedema associated with congestive heart failure, hepatic cirrhosis and renal disease (including nephrotic syndrome). Treatment of mild to moderate degree of hypertension.

Dosage and Administration(s):
For IM/IV use only.
For intravenous administration, furosemide must be injected or infused slowly; a rate of 4mg per minute must not be exceeded.
For intramuscular administration, it must be restricted to exceptional cases where neither oral nor intravenous administration is feasible. Intramuscular injection is not suitable for the treatment of acute conditions such as pulmonary oedema.

Adult:
For oedema, the usual initial dose is 1-2 amps once daily, adjusted as necessary according to response.
For hypertension, daily doses of 2-4 amps are given either alone or in conjunction with other antihypertensive.

Elderly:
In the elderly furosemide is generally eliminated more slowly. Dosage should be titrated until the required response is achieved.

Contraindication(s):
1. This product is contra-indicated in patients with hypovolaemia or dehydration, anuria or renal failure with anuria not responding to furosemide, renal failure as a result of poisoning by nephrotoxic or hepatotoxic agents or renal failure associated with hepatic coma, severe hypokalaemia, severe hyponatraemia, pre-comatose and comatose states associated with hepatic encephalopathy and breast feeding women.
2. Hypersensitivity to furosemide.
3. Patients allergic to sulphonamides may show cross-sensitivity to furosemide.

Precaution(s) / Warning(s):
1. Urinary output must be secured. Patients with partial obstruction of urinary outflow, for example patients with prostatic hypertrophy or impairment of micturition have an increased risk of developing acute retention and require careful monitoring.
2. Where indicated, steps should be taken to correct hypotension or hypovolaemia before commencing therapy.
3. Particularly careful monitoring is necessary in:
   - Patient with hypotension, risk from a pronounced fall in blood pressure, latent diabetes, gout, hepatorenal syndrome, hypoproteinaemia, e.g. associated with nephritic syndrome (the effect of furosemide may be weakened and its ototoxicity potentiated) and premature infants (possible development nephrocalcinosis/nephrolithiasis; renal function must be monitored and renal ultrasonography performed).
4. Caution should be observed in patients liable to electrolyte deficiency. Regular monitoring of serum sodium, potassium and creatinine is generally recommended during furosemide therapy; particularly close monitoring is required in patients at high risk of developing electrolyte imbalances or in case of significant additional fluid loss.
5. Hypovolaemia or dehydration as well as any significant electrolyte and acid-base disturbances must be corrected.This may require temporary discontinuation of furosemide.
6. In patients who are at high risk for radiocontrast nephropathy, furosemide is not recommended to be used for diuresis as part of the preventative measures against radiocontrast-induced nephropathy.

<u>Concomitant use with risperidone</u>
1. Concomitant use of furosemide and risperidone may cause a higher incidence of mortality. Concomitant use of risperidone with other diuretics (mainly thiazide diuretics used in low dose) was not associated with similar findings.
2. Caution should be exercised and the risks and benefits of this combination or co-treatment with other potent diuretics should be considered prior to the decision to use. Irrespective of treatment, dehydration was an overall risk factor for mortality and should therefore be avoided in elderly patients with dementia

Interaction with Other Medicaments:
1. The dosage of concurrently administered cardiac glycosides, diuretics, anti-hypertensive agents, or other drugs with blood-pressure-lowering potential may require adjustment as a more pronounced fall in blood pressure must be anticipated if given concomitantly with furosemide. A marked fall in blood pressure and deterioration in renal function may be seen when ACE inhibitors or angiotensin II receptor antagonists are added to furosemide therapy or their dose level increased. The dose of furosemide should be reduced for at least three days, or the drug stopped, before initiating the ACE inhibitor or angiotensin II receptor antagonist or increasing their dose.
2. The toxic effects of nephrotoxic drugs may be increased by concomitant administration of furosemide.
3. Oral furosemide and sucralfate must not be taken within 2 hours of each other because sucralfate decreases the absorption of furosemide.
4. Serum lithium levels may be increased when lithium is given concomitantly with furosemide, resulting in increased lithium toxicity, including increased risk of cardiotoxic and neurotoxic effects of lithium.
5. Risperidone: Caution should be exercised and the risks and benefits of the combination or co-treatment with furosemide or with other potent diuretics should be considered prior to the decision to use.

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ATTACHMENT 2

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6. Certain non-steroidal anti-inflammatory agents (e.g. indometacin, acetylsalicylic acid) may attenuate the action of furosemide and may cause acute renal failure in cases of pre-existing hypovolaemia or dehydration.
7. Furosemide may sometimes attenuate the effects of other drugs (e.g. the effects of anti-diabetics and of pressor amines) and sometimes potentiate them (e.g. the effects of salicylates, theophylline and curare-type muscle relaxants).
8. Furosemide may potentiate the ototoxicity of aminoglycosides and other ototoxic drugs. These drugs must only be used with furosemide if there are compelling medical reasons.
9. There is a risk of ototoxic effects if cisplatin and furosemide are given concomitantly. In addition, nephrotoxicity of cisplatin may be enhanced if furosemide is not given in low doses and with positive fluid balance when used to achieve forced diuresis during cisplatin treatment.
10. Attenuation of the effect of furosemide may occur following concurrent administration of phenytoin.
11. Concomitant administration of carbamazepine or aminoglutethimide may increase the risk of hyponatraemia.
12. Corticosteroids administered concurrently may cause sodium retention.
13. Corticosteroids, carbenoxolone, liquorice, B2 sympathomimetics in large amounts, prolonged use of laxatives, reboxetine and amphotericin may increase the risk of developing hypokalaemia.
14. Probenecid, methotrexate and other drugs which, like furosemide, undergo significant renal tubular secretion may reduce the effect of furosemide. Conversely, furosemide may decrease renal elimination of these drugs.
15. Impairment of renal function may develop in patients receiving concurrent treatment with furosemide and high doses of certain cephalosporins.
16. Concomitant use of ciclosporin and furosemide is associated with increased risk of gouty arthritis.

Pregnancy and Lactation:
1. There is clinical evidence of safety of the drug in the third trimester of human pregnancy; however, furosemide crosses the placental barrier. It must not be given during pregnancy unless there are compelling medical reasons. Treatment during pregnancy requires monitoring of foetal growth.
2. Furosemide passes into breast milk and may inhibit lactation. Women must not breast-feed if they are treated with furosemide.

Side Effect(s):
1. Eosinophilia, thrombocytopenia, leucopenia, agranulocytosis, aplastic anaemia or haemolytic anaemia may develop.
2. Bone marrow depression has been reported as a rare complication.
3. Paraesthesiae may occur.
4. Hepatic encephalopathy in patients with hepatocellular insufficiency may occur.
5. Serum calcium levels may be reduced; tetany has been observed.
6. Nephrocalcinosis / Nephrolithiasis has been reported in premature infants.
7. Serum cholesterol and triglyceride levels may rise during furosemide treatment.
8. Glucose tolerance may decrease with furosemide.
9. Hearing disorders and tinnitus.
10. Furosemide may cause a reduction in blood pressure.
11. Intrahepatic cholestasis, an increase in liver transaminases or acute pancreatitis may develop.
12. The incidence of allergic reactions, such as skin rashes, photosensitivity, vasculitis, fever, interstitial nephritis or shock is very low, but when these occur treatment should be withdrawn.
13. Furosemide leads to increased excretion of sodium and chloride and consequently water. Symptomatic electrolyte disturbances and metabolic alkalosis may develop in the form of a gradually increasing electrolyte deficit or, e.g. where higher furosemide doses are administered to patients with normal renal function, acute severe electrolyte losses. Warning signs of electrolyte disturbances include increased thirst, headache, hypotension, confusion, muscle cramps, tetany, muscle weakness, disorders of cardiac rhythm and gastrointestinal symptoms.
14. Pre-existing metabolic alkalosis (e.g. in decompensated cirrhosis of the liver) may be aggravated by furosemide treatment.
15. The diuretic action of furosemide may lead to or contribute to hypovolaemia and dehydration, especially in elderly patients. Severe fluid depletion may lead to haemoconcentration with a tendency for thromboses to develop.
16. Increased production of urine may provoke or aggravate complaints in patients with an obstruction of urinary outflow.
17. If furosemide is administered to premature infants during the first weeks of life, it may increase the risk of persistence of patent ductus arteriosus.
18. Severe anaphylactic or anaphylactoid reactions (e.g. with shock) occur rarely.
19. Side-effects of a minor nature such as nausea, malaise or gastric upset (vomiting or diarrhoea) may occur.
20. Following intramuscular injection, local reactions such as pain may occur.
21. Treatment with furosemide may lead to transitory increases in blood creatinine and urea levels. Serum levels of uric acid may increase and attacks of gout may occur.
22. Reduced mental alertness may impair ability to drive or operate dangerous machinery.

Symptoms and Treatment for Overdosage, and Antidote(s):
1. The clinical picture in acute or chronic overdose depends primarily on the extent and consequences of electrolyte and fluid loss, e.g. hypovolaemia, dehydration, haemoconcentration, cardiac arrhythmias due to excessive diuresis.
2. Symptoms of these disturbances include severe hypotension (progressing to shock), acute renal failure, thrombosis, delirious states, flaccid paralysis, apathy and confusion.
3. Treatment should therefore be aimed at fluid replacement and correction of the electrolyte imbalance. Together with the prevention and treatment of serious complications resulting from such disturbances and of other effects on the body, this corrective action may necessitate general and specific intensive medical monitoring and therapeutic measures.
4. No specific antidote to furosemide is known. If ingestion has only just taken place, attempts may be made to limit further systemic absorption of the active ingredient by measures such as gastric lavage or those designated to reduce absorption (e.g. activated charcoal).

Incompatibilities
Furosemide may precipitate out of solution in fluids of low pH (e.g. dextrose solutions).

Shelf-Life:
3 years

Storage Condition(s):
Store at temperature below 30°C

Product Description:
A clear and colorless to slightly yellowish solution.

Dosage Forms and Packaging available:
2ml x 10, 2ml x 50 and 2ml x 100 amber ampoules.

Y. S. P.®

Manufacturer and Product Registration Holder:
Y.S.P. INDUSTRIES (M) SDN. BHD. (199001001034)
Lot 3, 5, 7, 12 & 14, Jalan P/7, Section 13,
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Ordering Line: 1 800 88 3027
Product Info: 1 800 88 3679

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Date of revision: 24 Aug 2020

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